HW4

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Q1

Let $\underline{x}$ be an $n \times 1$ random vector with mean vector $\underline{\mu}$ and variance-covariance matrix Σ. Use Theorem 3.1 to show that $\underline{x}$ has a multivariate normal distribution if and only if, for every $n \times 1$ vector of constants $\underline{a}$, $\underline{a}'\underline{x}$ has a (univariate) normal distribution.

Answer

Q2

Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$. Show that

$$\operatorname{cov}(\underline{x}' A \underline{x}, \underline{x}' B \underline{x}) = 2 \operatorname{tr}(A\Sigma B\Sigma) + 4\underline{\mu}' A\Sigma B \underline{\mu},$$

where A and B are symmetric matrices of constants. [Hint: For any random variables w and z, $\operatorname{Var}(w+z) = \operatorname{Var}(w) + \operatorname{Var}(z) + 2\operatorname{cov}(w, z)$.]

Answer

Q3

Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is an $n \times 1$ vector and $\text{rank}(\Sigma) \geq 1$. Let

$$w_1 = \underline{b}_1'\underline{x} + \underline{x}'A_1\underline{x} \quad \text{and} \quad w_2 = \underline{b}_2'\underline{x} + \underline{x}'A_2\underline{x},$$

where $\underline{b}_1$ and $\underline{b}_2$ are vectors of constants and A_1 and A_2 are symmetric matrices of constants. Let

$$\underline{c}_1 = \underline{b}_1 + 2A_1\underline{\mu} \quad \text{and} \quad \underline{c}_2 = \underline{b}_2 + 2A_2\underline{\mu}.$$

(a)

Find

$$\text{cov}\left(\begin{pmatrix}\underline{b}_1' \\ A_1\end{pmatrix}\underline{x}, \begin{pmatrix}\underline{b}_2' \\ A_2\end{pmatrix}\underline{x}\right).$$

Answer

(b)

Suppose that (i) $\underline{b}_1'\Sigma\underline{b}_2 = 0$, (ii) $\underline{b}_1'\Sigma A_2 = \underline{0}'$, (iii) $\underline{b}_2'\Sigma A_1 = \underline{0}'$, and (iv) $A_1\Sigma A_2 = 0$. Use (a) to show that w_1 and w_2 are independent. [Hint: Note that $w_i = \underline{b}_i'\underline{x} + (A_i\underline{x})'A_i^-(A_i\underline{x})$ for $i = 1, 2$.]

Answer

Q4

Take the model to be Model II and assume that $\underline{y}$ has a multivariate normal distribution.

(a)

Obtain the expectation, the variance, and the mean squared error for any estimator of σ^2 having the form $(1/k)\underline{y}'(\underline{y} - X\widehat{\underline{\beta}})$, where k is a nonzero constant, and $\widehat{\underline{\beta}}$ is any solution to the normal equations.

Answer

(b)

Of those estimators of σ^2 having the above form, which has the smallest mean squared error? (That is, determine the optimum value of k.)

Answer

Q5

Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is an $r \times 1$ vector and $\text{rank}(\Sigma) = k \geq 1$ (where possibly $k < r$). Let A be an $r \times r$ symmetric matrix. Show that, if $A\Sigma$ is idempotent of rank s and if in addition $\underline{\mu}$ lies in $\mathcal{C}(\Sigma)$, then $\underline{x}'A\underline{x} \sim \chi^2\left(s, \frac{1}{2}\underline{\mu}'A\underline{\mu}\right)$. (Hint: Let Γ be a $k \times r$ matrix such that $\Sigma = \Gamma'\Gamma$, and show that $A\Sigma$ is idempotent of rank $s \implies \Gamma A\Gamma'$ is idempotent of rank s. Then, use $\underline{x}$ has the same distribution as $\Gamma'(\underline{z} + \underline{a})$, where $\underline{z}$ is a standard normal vector and $\underline{a}$ is a certain vector of constants.)

Answer

Q6

Let $\underline{x} \sim N(\underline{0}, I)$, and take A and B to be two symmetric matrices such that $\underline{x}'A\underline{x} \sim \chi^2(p)$ and $\underline{x}'B\underline{x} \sim \chi^2(q)$, where $p > q$. Show, by completing the following three steps, that $\underline{x}'(A-B)\underline{x} \sim \chi^2(p-q)$ if and only if the correlation $\text{corr}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = q/\sqrt{pq}$.

(a)

Show that $\text{tr}[(I-A)B] = 0$ if and only if $\text{corr}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = q/\sqrt{pq}$. [Hint: Consider $\text{cov}(\underline{x}'(A-B)\underline{x} + \underline{x}'B\underline{x}, \underline{x}'B\underline{x})$]

Answer

(b)

Show that $(I-A)B = 0$ if and only if $\text{tr}[(I-A)B] = 0$.

Answer

(c)

Show that $\underline{x}'(A-B)\underline{x} \sim \chi^2(p-q)$ if and only if $(I-A)B = 0$.

Answer